

Math 248 Midterm

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Problem 1. Let $n \in \mathbb{N}$ and I be a non-empty open interval in $\mathbb{R}$. Let $A : I \rightarrow M(n, \mathbb{C})$ and $B : M(n, \mathbb{C})$ be smooth paths.

1. Prove that $[A(t)]' = [A'(t)]^T$.

Proof. To show the above result, we need to show that

$$(A^T)'_{ij}(t) = (A')^T_{ij}(t) \quad \forall 1 \leq i, j \leq n.$$

Since $(A^T)_{ij} = A_{ji}$, we obtain that

$$\begin{aligned} (A^T)'_{ij} &= \lim_{\Delta t \rightarrow 0} \frac{(A^T)_{ij}(t + \Delta t) - (A^T)_{ij}}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{A_{ji}(t + \Delta t) - A_{ji}(t)}{\Delta t} \\ &= A'_{ji}(t) \\ &= (A'_{ij})^T(t) \end{aligned}$$

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2. Prove that $[A(t)B(t)]' = A'(t)B(t) + A(t)B'(t)$.

Proof. We have that

$$\begin{aligned} [A(t)B(t)]' &= \lim_{\Delta t \rightarrow 0} \frac{A(t + \Delta t)B(t + \Delta t) - A(t)B(t)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \left[B(t + \Delta t) \frac{A(t + \Delta t) - A(t)}{\Delta t} + A(t) \frac{B(t + \Delta t) - B(t)}{\Delta t} \right] \\ &= B(t) \lim_{\Delta t \rightarrow 0} \frac{A(t + \Delta t) - A(t)}{\Delta t} + A(t) \lim_{\Delta t \rightarrow 0} \frac{B(t + \Delta t) - B(t)}{\Delta t} \\ &= B(t)A'(t) + A(t)B'(t). \end{aligned}$$

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3. Prove that if, in addition, $A(t)$ is invertible for all t , then

$$[A^{-1}(t)]' = -A^{-1}(t)A'(t)A^{-1}(t).$$

Proof. Note that since $A(t)$ is invertible for all t , it follows that $AA^{-1} = I$. Using part (2), we have $[A(t)A^{-1}(t)]' = 0$. Differentiating, we see that

$$\begin{aligned} A'(t)A^{-1}(t) + A(t)[A^{-1}(t)]' &= 0 \implies A(t)[A^{-1}(t)]' = -A'(t)A^{-1}(t) \\ &\implies (A^{-1})'(t) = A^{-1}(t)(-A'(t))A^{-1}(t). \end{aligned}$$

That is,

$$\begin{aligned}(A^{-1})'(t) &= A^{-1}(t)(-A'(t))A^{-1}(t) \\ &= -A^{-1}(t)A'(t)A^{-1}(t).\end{aligned}$$

Hence, we conclude that

$$(A^{-1})'(t) = -A^{-1}(t)A'(t)A^{-1}(t).$$

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- Problem 2.**
1. Explain why $E = [3, 5]$ is a closed and bounded subset of the metric space $X = (1, 5)$.
 2. Explain why $E = [3, 5]$ is not a compact subset of the metric space $X = (1, 5)$.
 3. Explain why the following argument is faulty. " $O(n)$ is compact because it is a closed subset of $GL(n, \mathbb{R})$ and is bounded."

- Solution.**
1. E is closed since $[3, 5] = [3, 6] \cap (1, 5)$ where $[3, 6]$ is closed and bounded set in $\mathbb{R}$.
 2. $E = [3, 5]$ is not compact in X because E is NOT closed in $\mathbb{R}$ (that is, $5 \notin E$). Hence, E cannot be compact in $X = (1, 5)$ by fact 1.
 3. This argument is faulty because in general if $X = \mathbb{R}^k$, then we cannot guarantee that a set being closed and bounded in X will be compact in X . In particular, $O(n)$ being closed and bounded in $GL(n, \mathbb{R})$ does not imply that it is a compact set in $GL(n, \mathbb{R})$ since $GL(n, \mathbb{R}) \neq \mathbb{R}^k$. Furthermore, $GL(n, \mathbb{R})$ is not even a compoact set in $\mathbb{R}^{n^2}$ as we have proven in class and so the fact

"closed subsets of compact sets are compact"

cannot be used to make such a statement.

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Problem 3. Let V be an n -dimensional vector space. Suppose $T : V \rightarrow V$ is a linear transformation. As we know, once we equip V with a basis, we can represent T by a matrix. Let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V . Denote the matrix of T relative to the basis $\mathcal{B}$ by $A_{\mathcal{B}}$ and the matrix of T relative to the basis $\mathcal{C}$ by $A_{\mathcal{C}}$. Thus, for every $\mathbf{x} \in V$, we have

$$\begin{aligned}[T(\mathbf{x})]_{\mathcal{B}} &= A_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}, \\ [T(\mathbf{x})]_{\mathcal{C}} &= A_{\mathcal{C}}[\mathbf{x}]_{\mathcal{C}}.\end{aligned}$$

Prove that

$$A_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} A_{\mathcal{B}} (P_{\mathcal{C} \leftarrow \mathcal{B}})^{-1}.$$

Proof. Let $[\mathbf{x}]_{\mathcal{C}} \in \mathbb{R}^n$ be arbitrary. Then we have

$$\begin{aligned}P_{\mathcal{C} \leftarrow \mathcal{B}} A_{\mathcal{B}} ((P_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} [\mathbf{x}]_{\mathcal{C}}) &= P_{\mathcal{C} \leftarrow \mathcal{B}} A_{\mathcal{B}} (P_{\mathcal{B} \leftarrow \mathcal{C}} [\mathbf{x}]_{\mathcal{C}}) \\ &= P_{\mathcal{C} \leftarrow \mathcal{B}} (A_{\mathcal{B}} [\mathbf{x}]_{\mathcal{B}}) \\ &= P_{\mathcal{C} \leftarrow \mathcal{B}} [T(\mathbf{x})]_{\mathcal{B}} \\ &= [T(\mathbf{x})]_{\mathcal{C}} \\ &= A_{\mathcal{C}} [\mathbf{x}]_{\mathcal{C}}.\end{aligned}$$

Hence, we have

$$P_{\mathcal{C} \leftarrow \mathcal{B}} A_{\mathcal{B}} ((P_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} [\mathbf{x}]_{\mathcal{C}}) = A_{\mathcal{C}} [\mathbf{x}]_{\mathcal{C}}.$$

Multiplying $[\mathbf{x}]_C^{-1}$ on both sides, we obtain that

$$\mathcal{A}_C = P_{C \leftarrow B} \mathcal{A}_B (P_{C \leftarrow B})^{-1}.$$

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Problem 4. Let $N = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

1. Prove that N is nilpotent.

Proof. We have

$$\begin{aligned} N^2 &= \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^2 \\ &= \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= O. \end{aligned}$$

Thus, N is nilpotent with $k = 2$ such that $N^2 = O$.

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2. Compute $\exp(N)$.

Solution. Note that for all $k \geq 2$, $A^k = O$. Using the definition of the matrix exponential, we obtain

$$\begin{aligned} \exp(N) &= \sum_{m=0}^{\infty} \frac{N^m}{m!} = \frac{N^0}{0!} + \frac{N^1}{1!} + \frac{N^2}{2!} + 0 + 0 + \dots \\ &= I + N \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}. \end{aligned}$$

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Problem 5. Let $\theta \in \mathbb{R}$. Find the exponential of the matrix $X = \begin{pmatrix} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Solution. Let $\lambda_1 = i\theta$, $\lambda_2 = -i\theta$ and $\lambda_3 = 0$. Observe that

$$\begin{aligned} x \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \begin{pmatrix} -i\theta \\ \theta \\ 0 \end{pmatrix} = i\theta \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \\ x \begin{pmatrix} i \\ 1 \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} i \\ 1 \\ 0 \end{pmatrix} = -i\theta \begin{pmatrix} -\theta \\ i\theta \\ 0 \end{pmatrix} = -i\theta \begin{pmatrix} i \\ 1 \\ 0 \end{pmatrix} \\ x \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned}$$

Then we see that $P = \begin{pmatrix} 1 & i & 0 \\ i & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $P^{-1} = \begin{pmatrix} 1/2 & i/2 & 0 \\ -i/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Furthermore, we obtain

$$\begin{aligned} P^{-1}DP &= \begin{pmatrix} 1 & i & 0 \\ i & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -i\theta & 0 & 0 \\ 0 & i\theta & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1/2 & i/2 & 0 \\ -i/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -i\theta & -\theta & 0 \\ \theta & i\theta & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1/2 & -i/2 & 0 \\ -i/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= X. \end{aligned}$$

Hence, we have

$$X = P^{-1}DP \text{ where } D = \begin{pmatrix} -i\theta & 0 & 0 \\ 0 & i\theta & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since $e^X = Pe^DP^{-1}$, we obtain that

$$\begin{aligned} e^X &= Pe^DP^{-1} \\ &= \begin{pmatrix} i & 1 & 0 \\ 1 & i & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} e^{-i\theta} & 0 & 0 \\ 0 & e^{i\theta} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1/2 & -i/2 & 0 \\ -i/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} ie^{-i\theta} & e^{i\theta} & 0 \\ e^{-i\theta} & ie^{i\theta} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1/2 & -i/2 & 0 \\ -i/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{e^{-i\theta} - e^{i\theta}}{2} & \frac{-ie^{-i\theta} + ie^{i\theta}}{2} & 0 \\ \frac{ie^{-i\theta} - ie^{i\theta}}{2} & \frac{e^{-i\theta} + e^{i\theta}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{e^{-i\theta} - e^{i\theta}}{2i} & \frac{-(ie^{-i\theta} - e^{i\theta})}{2} & 0 \\ \frac{e^{i\theta} - e^{-i\theta}}{2i} & \frac{e^{-i\theta} + e^{i\theta}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

Hence, we have that

$$e^X = Pe^D P^{-1} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

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Problem 6. Let $A \in O(n)$ and $B \in M(n, \mathbb{R})$. Prove that $\exp(ABA^T) = A \exp(B) A^T$.

Proof. One can prove via induction that $\exp(ABA^T) = \sum_{m=0}^{\infty} \frac{(ABA^T)^m}{m!}$. Using this fact, we obtain

$$\begin{aligned} \exp(ABA^T) &= \sum_{m=0}^{\infty} \frac{(ABA^T)^m}{m!} = \sum_{m=0}^{\infty} \frac{AB^m A^T}{m!} \\ &= \sum_{m=0}^{\infty} A \frac{B^m}{m!} A^T \\ &= A \sum_{m=0}^{\infty} \frac{B^m}{m!} A^T \\ &= A \exp(B) A^T. \end{aligned}$$

Now, we conclude that

$$\exp(ABA^T) = A \exp(B) A^T.$$

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Problem 7. Let G be any of the matrix Lie groups that we have introduced as embedded submanifolds of $\text{GL}(n, \mathbb{F})$. Prove that $T_I G$ is a linear subspace of $\mathbb{R}^k$ where $k = n^2$ if $\mathbb{F} = \mathbb{R}$ and $k = 2n^2$ if $\mathbb{F} = \mathbb{C}$.

Proof. It suffices to show that $T_I G$ is closed under addition and scalar multiplication.

(i) $T_I G$ is closed under addition.

Let $X, Y \in T_I G$. Then there exists smooth paths $f_1(t), f_2(t) \in G$ such that $f_1(0) = I$, $f_1'(0) = X$, $f_2(0) = I$, and $f_2'(0) = Y$. Define $f_3(t) = f_1(t)f_2(t)$. Then notice that

$$f_3'(t) = f_1'(t)f_2(t) + f_2'(t)f_1(t)$$

and so

$$\begin{aligned} f_3'(0) &= f_1'(0)f_2(0) + f_2'(0)f_1(0) \\ &= XI + YI \\ &= X + Y. \end{aligned}$$

Thus, $f_3'(0) = X + Y$ and clearly $f_3(0) = I$ since

$$f_3(0) = f_1(0)f_2(0) = II = I.$$

Therefore, we have $X + Y \in T_I G$.

(ii) $T_I G$ is closed under scalar multiplication.

Let $\alpha \in \mathbb{F}$ and let $X \in T_I G$. Then there exists a smooth path $f(t)$ in G such that $f(0) = I$ and $f'(0) = X$. Define $g(t) = f(\alpha t)$. Using the chain rule, we get

$$g'(t) = f'(\alpha t) \cdot \alpha \implies g'(0) = \alpha \cdot f'(0) = \alpha X$$

and $g(0) = f(\alpha \cdot 0) = f(0) = I$. Hence, we see that $\alpha X \in T_I G$.

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Problem 8. Let $\mathbb{F}$ denote either $\mathbb{R}$ or $\mathbb{C}$. Prove that the *commutator bracket* $[A, B] = AB - BA$ defines a Lie Bracket in the vector space $M(n, \mathbb{F})$. (Thus, $M(n, \mathbb{F})$ equipped with this Lie Bracket is a Lie algebra, often denoted by $\mathfrak{gl}(n, \mathbb{F})$).

Proof. (i) Let $\alpha, \lambda \in \mathbb{F}$ and let $A, B, C \in M(n, \mathbb{F})$. Then

$$\begin{aligned} [\alpha A + \lambda B, C] &= (\alpha A + \lambda B)C - C(\alpha A + \lambda B) \\ &= \alpha(AC) + \lambda(BC) - C(\alpha A) - C(\lambda B) \\ &= \alpha(AC) + \lambda(BC) - \alpha(CA) - \lambda(CB) \\ &= \alpha[AC - CA] + \lambda[BC - CB] \\ &= \alpha[A, C] + \lambda[B, C] \end{aligned}$$

Thus, we obtain

$$[\alpha A + \lambda B, C] = \alpha[A, C] + \lambda[B, C].$$

A similar argument shows linearity in the second component.

(ii) Observe that

$$[A, B] = AB - BA = -(BA - AB) = -[B, A].$$

(iii) Let $X, Y, Z \in M(n, \mathbb{F})$. Note that

$$\begin{aligned} [X, [Y, Z]] &= X[Y, Z] - [Y, Z]X \\ &= X(YZ - ZY) - (YZ - ZY)X \\ &= X(YZ) - X(ZY) - (YZ)X + (ZY)X \end{aligned}$$

Then we have

$$\begin{aligned} [Y, [Z, X]] &= Y(ZX) - Y(XZ) - (ZX)Y + (XZ)Y \\ [Z, [X, Y]] &= Z(XY) - Z(YX) - (XY)Z + (YX)Z. \end{aligned}$$

Then we have

$$\begin{aligned} &[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] \\ &= X(YZ) - (XY)Z + (YX)Z - Y(XZ) + (XZ)Y - X(ZY) \\ &\quad + Y(ZX) - (YZ)X + (ZY)X - Z(YX) + (ZX)Y - (ZX)Y \\ &= 0. \end{aligned}$$

Hence, $[\cdot, \cdot]$ satisfies the Jacobi Identity. Thus, we conclude that $[A, B] = AB - BA$ defines a Lie Bracket on $M(n, \mathbb{F})$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. ■

Problem 9. Let V be a finite-dimensional real or complex vector space. Let $\text{End}(V)$ denote the vector space of all linear transformations from V to V . Prove that the *commutator bracket* $[f, g] = f \circ g - g \circ f$ defines a Lie Bracket on the vector space $\text{End}(V)$. (Thus, $\text{End}(V)$ equipped with this Lie Bracket is a Lie Algebra, often denoted by $\mathfrak{gl}(V)$ and called the *general linear algebra of V* .)

Proof. Let $\alpha, \beta \in \mathbb{F}$ and let $f, g, h \in \text{End}(V)$.

(i) (Bilinearity) For all $x \in V$, we have

$$\begin{aligned}
[\alpha f + \beta g, h](x) &= (\alpha f + \beta g) \circ h(x) - (h \circ \alpha f + \beta g)(x) \\
&= (\alpha f + \beta g)(h(x)) - h((\alpha f + \beta g)(x)) \\
&= \alpha f(h(x)) + \beta g(h(x)) - h((\alpha f)(x) + (\beta g)(x)) \\
&= \alpha f(h(x)) + \beta g(h(x)) - h(\alpha f(x)) - h(\beta g(x)) \\
&= \alpha f(h(x)) + \beta g(h(x)) - \alpha h(f(x)) - \beta h(g(x)) \\
&= \alpha(f \circ h)(x) + \beta(g \circ h)(x) - \alpha(h \circ f)(x) - \beta(h \circ g)(x) \\
&= \alpha[f \circ h - h \circ f](x) + \beta[g \circ h - h \circ g](x) \\
&= \alpha[f, h](x) + \beta[g, h](x).
\end{aligned}$$

Hence, we conclude that

$$[\alpha f, \beta g, h] = \alpha[f, h] + \beta[g, h].$$

A similar argument shows bilinearity for the second component.

(ii) (Skew-symmetric) Let $f, g \in \text{End}(V)$. Then

$$[f, g] = f \circ g - g \circ f = -(g \circ f - f \circ g) = -[g, f].$$

(iii) Let $f, g, h \in \text{End}(V)$. Our goal is to show that

$$[f, [g, h]] + [h, [f, g]] + [g, [h, f]] = 0$$

Note that

$$\begin{aligned}
[f, [g, h]] &= f \circ [g, h] - [g, h] \circ f \\
&= f \circ (g \circ h - h \circ g) - (g \circ h - h \circ g) \circ f \\
&= f \circ (g \circ h) - f \circ (h \circ g) - (g \circ h) \circ f + (h \circ g) \circ f.
\end{aligned}$$

Similarly, we have that

$$\begin{aligned}
[h, [f, g]] &= h \circ [f, g] - [f, g] \circ h \\
&= h \circ [f \circ g - g \circ f] - [f \circ g - g \circ f] \circ h \\
&= h \circ (f \circ g) - h \circ (g \circ f) - (f \circ g) \circ h + (g \circ f) \circ h
\end{aligned}$$

and

$$\begin{aligned}
[g, [h, f]] &= g \circ [h, f] - [h, f] \circ g \\
&= g \circ [h \circ f - f \circ h] - [h \circ f - f \circ h] \circ g \\
&= g \circ (h \circ f) - g \circ (f \circ h) - (h \circ f) \circ g + (f \circ h) \circ g.
\end{aligned}$$

Then cancelling like terms and collecting, we arrive at

$$[f, [g, h]] + [h, [f, g]] + [g, [h, f]] = 0.$$

Hence, we conclude from above that $[f, g] = f \circ g - g \circ f$ is a Lie Bracket on $\text{End}(V)$. ■

Problem 10. Let G be any of the matrix Lie groups that we have introduced as embedded submanifolds of $\text{GL}(n, \mathbb{F})$. We know that $T_I G$ is a vector space of matrices. Prove that if $X, Y \in T_I G$, then $[X, Y] = XY - YX \in T_I G$ and hence $[X, Y] = XY - YX$ is a Lie Bracket on $T_I G$. (The space $T_I G$ equipped with this Lie Bracket is the Lie Algebra of G , denoted by $\text{Lie}(G)$ or $\mathfrak{g}$).

Proof. Suppose $X, Y \in T_I G$. Our goal is to show that $[X, Y] \in T_I G$. Since $X, Y \in T_I G$, there exists

smooth paths $A(t)$ and $B(t)$ such that

$$A(0) = I \quad \text{and} \quad B(0) = I$$

and

$$A'(0) = X \quad \text{and} \quad B'(0) = Y.$$

Define $C_S(t) = A(s)B(t)A^{-1}(S)$ is also a smooth path in $T_I G$. Hence, we see that for all s , $C'_s(0) = T_I G$. Now, if we let $V(s) = C'_s(0)$, then $V(s)$ is a path inside $T_I G$. Thus, in particular $V'(0) = T_I G$. That is,

$$\begin{aligned} \frac{d}{ds} \Big|_{s=0} \left[\frac{\partial}{\partial t} \Big|_{t=0} A(s)B(t)A^{-1}(s) \right] &\in T_I G \\ &= \frac{d}{ds} \Big|_{s=0} \left[A(s)B'(0)A^{-1}(s) \right] \\ &= A'(0)B'(0)A^{-1}(0) + A(0)B'(0)(A^{-1}(0))' \\ &= A'(0)B'(0)A^{-1}(0) + A(0)B'(0)[-A(0)A'(0)A^{-1}(0)] \\ &= XYI + IY[-IXI^{-1}] \\ &= XY - YX \\ &= [X, Y]. \end{aligned}$$

Hence, it follows that $[X, Y] \in T_I G$. ■

Problem 11. Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T(x, y, z) = (2x, x + 2y - z, x + 3y - 2z).$$

1. Find the matrix of T with respect to the standard basis $\mathcal{E} = \{e_1, e_2, e_3\}$ of $\mathbb{R}^3$.
2. Verify that the following vectors are eigenvectors of T :

$$v_1 = (1, 1, 1), \quad v_2 = (0, 1, 1) \quad v_3 = (0, 1, 3).$$

3. Let $\mathcal{B} = \{v_1, v_2, v_3\}$ be this basis of eigenvectors. Compute the matrix of T with respect to $\mathcal{B}$ and verify that is diagonal. (**Hint:** Use $[T]_{\mathcal{B}} = P^{-1}[T]_{\mathcal{E}}P$ with the change of basis matrix $P = P_{\mathcal{E} \leftarrow \mathcal{B}} = \begin{pmatrix} v_1 & v_2 & v_3 \end{pmatrix}$).

Solution. 1. Given $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$, we have

$$\begin{aligned} (x, y, z) &= xe_1 + ye_2 + ze_3 \\ \implies T(x, y, z) &= 2xe_1 + (x + 2y - z)e_2 + (x + 3y - 2z)e_3. \end{aligned}$$

Furthermore, we have

$$\begin{aligned} T(e_1) &= T(1, 0, 0) = (2, 1, 1) \\ T(e_2) &= T(0, 1, 0) = (0, 2, 3) \\ T(e_3) &= T(0, 0, 1) = (0, -1, -2). \end{aligned}$$

Using the above, we obtain the matrix representation of T with respect to $\mathcal{E}$:

$$[T]_{\mathcal{E}} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{pmatrix}.$$

2. Notice that

$$\begin{aligned} T(v_1) &= (2, 1 + 2(1) - 1, 1 + 3(1) - 2(1)) \\ &= (2, 2, 2) \\ &= 2v_1 \end{aligned}$$

with $\lambda_1 = 2$. Similarly, we get $T(v_2) = 1 \cdot v_2$ and $T(v_3) = -1 \cdot v_3$ where $\lambda_2 = 1$ and $\lambda_3 = -1$, respectively. Hence, the vectors listed above are, indeed, eigenvectors of T .

3. Using the eigenvectors above, we can construct the matrix representation of T with respect to the basis $\mathcal{E}$. That is, we arrive at

$$[T]_{\mathcal{E}} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{pmatrix}.$$

Note that the column vectors of P are just the eigenvectors from the previous part. Hence, we obtain that

$$P_{\mathcal{E} \leftarrow \mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

One can use the many computational techniques available to compute the inverse of $P_{\mathcal{E} \leftarrow \mathcal{B}}$. Indeed, we get (after using a computational software such as wolfram)

$$P_{\mathcal{E} \leftarrow \mathcal{B}}^{-1} [T]_{\mathcal{E}} P_{\mathcal{E} \leftarrow \mathcal{B}} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

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Problem 12. Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T(x, y, z) = (y, 0, x + y).$$

1. Verify that T is nilpotent, that is, there exists a $k > 0$ such that $T^k = 0$.

Solution. Using the definition of T , we get

$$T^2(x, y, z) = T(y, 0, x + y) = (0, 0, y + y) = (0, 0, 2y)$$

and

$$T^3(x, y, z) = T(0, 0, 2y) = (0, 0, 0)$$

Hence, T is nilpotent with $k = 3$ such that $T^k = 0$.

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2. Find the matrix of T with respect to the standard basis $\mathcal{E} = \{e_1, e_2, e_3\}$ of $\mathbb{R}^3$.

Solution. Observe that

$$T(e_1) = T(1, 0, 0) = (0, 0, 1) = e_3$$

$$T(e_2) = T(0, 1, 0) = (1, 0, 1)$$

$$T(e_3) = T(0, 0, 1) = (0, 0, 0).$$

Thus, we obtain the matrix representation of T with respect to $\mathcal{E}$

$$[T]_{\mathcal{E}} = (T(e_1) \quad T(e_2) \quad T(e_3)) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

■

3. Consider the new basis

$$\mathcal{B} = \{v_1 = e_3, \quad v_2 = e_1 + e_3, \quad v_3 = e_2\}$$

Solution. Note

$$P = P_{\mathcal{E} \leftarrow \mathcal{B}} = (v_1 \quad v_2 \quad v_3) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

and

$$P^{-1} = P_{\mathcal{E} \leftarrow \mathcal{B}}^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

To make our computations much easier, we will use software to get our final result. Indeed, we obtain using the change of coordinate matrices above

$$P^{-1}[T]_{\mathcal{E}}P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

■

Problem 13. Consider the linear transformation $T : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ defined by

$$T(x, y, z) = (3x - y + z, 2x, -x + y + 3z).$$

1. Find the matrix of T with respect to the standard basis $\mathcal{E} = \{e_1, e_2, e_3\}$ of $\mathbb{C}^3$.

Solution. Using the definition of T , we obtain

$$T(e_1) = T(1, 0, 0) = (3, 2, -1)$$

$$T(e_2) = T(0, 1, 0) = (-1, 0, 1)$$

$$T(e_3) = T(0, 0, 1) = (1, 0, 3).$$

Now, the matrix representation of T with respect to $\mathcal{E}$ is

$$[T]_{\mathcal{E}} = (T(e_1) \quad T(e_2) \quad T(e_3)) = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 0 \\ -1 & 1 & 3 \end{pmatrix}.$$

■

2. Consider the new basis

$$\mathcal{B} = \{v_1 = e_1 + e_2, \quad v_2 = e_1 - e_2 + 2e_3, \quad v_3 = -2e_1 + 2e_2 + e_3\}.$$

Compute the matrix of T with respect to $\mathcal{B}$ and verify that it is upper triangular.

Solution. Observe that $v_1 = (1, 1, 0)$, $v_2 = (1, -1, 2)$, and $v_3 = (-2, 2, 1)$. Define

$$P_{\mathcal{E} \leftarrow \mathcal{B}}$$

Lemma. For any $v \in L$ where L is a Lie Algebra, $[v, v] = 0$.

Proof. Let $v \in L$. By the Skew-symmetric property of $[\cdot, \cdot]$, it follows that

$$[v, v] = -[v, v] \implies 2[v, v] = 0 \implies [v, v] = 0.$$

Problem 14. A Lie Algebra is said to be *abelian* if

$$\forall x, y \in L \quad [x, y] = 0.$$

Prove that every one-dimensional Lie Algebra is abelian.

Proof. Let L be a one-dimensional Lie Algebra. Our goal is to show that for all $x, y \in L$, $[x, y] = 0$. Let $x, y \in L$ be arbitrary. Since L is a one-dimensional Lie Algebra, L is a one-dimensional vector space. Let $\beta = \{v\}$ be a basis for L . Then $x = \alpha_1 v$ and $y = \alpha_2 v$ for some α_1 and α_2 in $\mathbb{F}$. Using the scaling property of the first and second component of the Lie Bracket, it follows that

$$[x, y] = [\alpha_1 v, \alpha_2 v] = \alpha_1 \alpha_2 [v, v] = \alpha_1 \alpha_2 \cdot 0 = 0.$$

Hence, we see that L is abelian.

Problem 15. Recall that a Lie Algebra is a set equipped with three extra structures: addition and scalar multiplication (giving it the structure of a vector space), and a bilinear operation called the Lie Bracket. With this in mind, how should one define a *Lie Algebra Homomorphism* – a structure-preserving map between two Lie Algebras?

Definition (Lie Algebra Homomorphism). Let $\mathfrak{g}$ and $\mathfrak{h}$ be two Lie Algebras over $\mathbb{F}$ with Lie Brackets $[\cdot, \cdot]$ and $[\cdot, \cdot]'$, respectively. We say that the mapping $\varphi : \mathfrak{g} \rightarrow \mathfrak{h}$ is a **Lie Algebra Homomorphism** if φ is a linear map (i.e for all $x, y \in \mathfrak{g}$, for all $\alpha \in \mathbb{F}$ $\varphi(\alpha x + y) = \alpha \varphi(x) + \varphi(y)$.) such that

$$\varphi([x, y]) = [\varphi(x), \varphi(y)]' \quad \forall x, y \in \mathfrak{g}.$$

Problem 16. Let L be a real or complex Lie Algebra. For each $x \in L$, define a linear map:

$$\text{ad}_x : L \rightarrow L \quad \text{ad}_x(y) = [x, y].$$

1. Show that for every $x \in L$, the map $\text{ad}_x : L \rightarrow L$ is linear.

Proof. Let $x \in L$. Then by Bilinearity of $[\cdot, \cdot]$, we have for all $y_1, y_2 \in L$, we have

$$\begin{aligned} \text{ad}_x(y_1 + y_2) &= [x, y_1 + y_2] \\ &= [x, y_1] + [x, y_2] \\ &= \text{ad}_x(y_1) + \text{ad}_x(y_2). \end{aligned}$$

Hence, we have

$$\text{ad}_x(y_1 + y_2) = \text{ad}_x(y_1) + \text{ad}_x(y_2). \tag{1}$$

Now, let $\alpha \in \mathbb{F}$. Then for all $y \in L$, we have

$$\text{ad}_x(\alpha y) = [x, \alpha y] = \alpha [x, y] = \alpha \text{ad}_x(y).$$

Thus,

$$\text{ad}_x(\alpha y) = \alpha \cdot \text{ad}_x(y). \quad (2)$$

Now, it follows from (1) and (2) that $\text{ad}_x : L \rightarrow L$ is linear. ■

2. Define $\text{ad} : L \rightarrow \mathfrak{gl}(L)$ $x \mapsto \text{ad}_x$. Prove that ad is a linear map and that it preserves the Lie Bracket

$$\forall x, y \in L \quad [\text{ad}_x, \text{ad}_y] = \text{ad}_{[x, y]},$$

where the bracket on the left-hand side is the commutator bracket in $\mathfrak{gl}(L)$, and the bracket on the right-hand side is the Lie Bracket of L .

Proof. We will prove the following properties:

(1) $\text{ad} : L \rightarrow \mathfrak{gl}(L)$ defined by $x \mapsto \text{ad}_x$ is linear

(2) $\forall x, y \in L \quad [\text{ad}_x, \text{ad}_y] = \text{ad}_{[x, y]}$.

(1) Let $x, y \in L$. Then for all $z \in L$, we have

$$\begin{aligned} [\text{ad}(x + y)](z) &= \text{ad}_{x+y}(z) = [x + y, z] \\ &= [x, z] + [y, z] \\ &= \text{ad}_x(z) + \text{ad}_y(z). \end{aligned}$$

Then we see that

$$\text{ad}(x + y) = \text{ad}(x) + \text{ad}(y)$$

since $z \in L$ is arbitrary. Now, let $\alpha \in \mathbb{F}$. Then we have for all $z \in L$,

$$\begin{aligned} [\text{ad}(\alpha x)](z) &= \text{ad}_{\alpha x}(z) = [\alpha x, z] = \alpha[x, z] \\ &= \alpha \text{ad}_x(z). \end{aligned}$$

Hence, we see that $\text{ad} : L \rightarrow \mathfrak{gl}(L)$ is linear.

(2) Let $x, y \in L$. Then for all $z \in L$, it follows that

$$\begin{aligned} [\text{ad}_x, \text{ad}_y](z) &= (\text{ad}_x \circ \text{ad}_y - \text{ad}_y \circ \text{ad}_x)(z) \\ &= (\text{ad}_x \circ \text{ad}_y)(z) - (\text{ad}_y \circ \text{ad}_x)(z) \\ &= \text{ad}_x([y, z]) - \text{ad}_y([x, z]) \\ &= [x, [y, z]] - [y, [x, z]]. \end{aligned}$$

It suffices to show that

$$\text{ad}_{[x, y]}(z) = [[x, y], z] - [[x, y], z].$$

We proceed to use the Jacobi-identity. That is,

$$\begin{aligned} [y, [x, z]] + [z, [y, x]] + [x, [z, y]] &= 0 \\ \implies [z, [y, x]] + [x, [z, y]] &= -[y, [x, z]] \\ \implies [z, -[x, y]] + [x, [z, y]] &= -[y, [x, z]] \\ \implies -[-[x, y], z] + [x, [z, y]] &= -[y, [x, z]] \\ \implies [[x, y], z] + [x, -[y, z]] &= -[y, [x, z]] \\ \implies [[x, y], z] - [x, [y, z]] &= -[y, [x, z]]. \end{aligned}$$

Thus, we have that for all $z \in L$,

$$\begin{aligned} [\text{ad}_x, \text{ad}_y]k &= [x, [y, z]] + [[x, y], z] - [x, [y, z]] \\ &= [[x, y], z] \\ &= \text{ad}_{[x, y]}(z) \end{aligned}$$

Thus, we conclude that

$$[\mathrm{ad}_x, \mathrm{ad}_y] = \mathrm{ad}_{[x,y]}(z).$$

■